

EndoTracker: Robustly Tracking Any Point in Endoscopic Surgical Scene

Lalithkumar Seenivasan^{1,*}, Joanna Cheng^{1,*}, Jin Fang^{1,2,*}, Jin Bai¹, Roger D. Soberanis-Mukul¹, Jan Emily Mangulabnan¹, S. Swaroop Vedula¹, Masaru Ishii³, Gregory Hager¹, Russell H. Taylor¹, and Mathias Unberath¹

¹ Johns Hopkins University, Baltimore MD, USA

² Hefei University of Technology, Hefei, Anhui, China

³ Johns Hopkins Medical Institutions, Baltimore MD, USA
{lseeniv1,unberath}@jhu.edu

Abstract. The most exciting frontiers of contemporary endoscopic video processing include tissue and instrument tracking and scene reconstruction together with their applications in surgical navigation, mixed reality visualization, and surgical automation. These tasks are enabled by task-specific techniques such as segmentation, object tracking, structure from motion, simultaneous localization and mapping (SfM and SLAM), and neural rendering. While these techniques vary in purpose and methodology, fundamentally, they rely on the ability to reliably and robustly establish point correspondences across video frames. However, point tracking in endoscopic scenes is hard due to the lack of distinct yet repetitive features, varying illuminations, and continuously changing visual appearance of corresponding points. A dense point-tracking model capable of reliably establishing point correspondences across video frames could catalyze endoscopic video processing and its downstream applications, and drive significant advancement in surgical data science. While any point-tracking foundation models have recently been proposed, they are trained on large simulated and natural scene videos and need to be adapted to endoscopic scenes to address domain-specific challenges. In this work, we present a dense point tracking foundation model for endoscopic scenes by fine-tuning a public foundation model on a large custom dataset comprising 13k endoscopic video sequences (314k frames). We present three benchmark datasets with ground-truth point correspondences to quantitatively evaluate the point tracking performance in variable endoscopic scenes. Through quantitative analysis, we find that models fine-tuned on endoscopic scenes outperform out-of-the-box models, especially when considering conservative thresholds for tracking success, suggesting improved suitability for downstream tasks due to error propagation.

Keywords: Tracking Any Point · Foundation Model · Deep Learning

1 Introduction

Tissue and instrument tracking, and scene reconstruction are at the forefront of endoscopic video processing, spearheading the transformation in surgical navi-

gation [21], mixed reality visualization [16], and surgical automation [3]. These tasks are currently enabled by contemporary task-specific techniques that include segmentation [19, 4], object detection [9], tool-tip detection [17], structure from motion (SfM), and simultaneous localization and mapping [20]. Despite their varied objectives and approaches, these techniques fundamentally rely on accurately and consistently establishing temporal point correspondences across video frames. Methods addressing these fundamental needs – any point-tracking – could potentially revolutionize the approaches in enabling a wide range of downstream tasks in the surgical domain. The introduction of any point tracking foundation models in the computer vision domain, such as PIPs [11], TAPIR [8], and CoTracker [14, 13], has initiated its rapid adoption with the surgical domain, with downstream tasks focusing on tracking instrument and tissues [22], and surgical navigation [20]. While these point tracking models show great promise as a key enabler for surgical applications, their adoption for clinical applications is severely hindered by domain shifts and domain-specific challenges.

The introduction of the TAPVid large-scale dataset [7] and Kubric synthetic dataset [10] has catalyzed the advancement of any point tracking foundation models [8, 14, 13]. Trained on massive synthetic datasets, these models allow tracking of any points in natural dynamic scenes. Although they exhibit generalizability to the surgical domain to an extent, with recent works adopting these models to aid in instrument tracking [22] and surgical navigation [20], they are severely limited by domain shift and domain-specific challenges. Firstly, unlike natural scenes, tissue surfaces exhibit homogeneous visual characteristics, often lacking distinct features for the model to reason and establish accurate point correspondences across frames. Secondly, surgical scenes suffer from temporal photometric inconsistency due to varying lighting conditions, causing the same tissue surfaces to exhibit different textures and reflections across frames.

To this end, we first custom-curate a large endoscopic dataset - comprising of 13k short endoscopic video sequences (each consisting of 24 frames) - to fine-tune any-point tracking foundation models in a self-supervised setting. To quantitatively evaluate point-tracking models on endoscopic scenes, we additionally introduce 3 point-tracking benchmarking datasets. We further present EndoTracker—an adaptation of CoTracker3 [13] fine-tuned on our meticulously curated large endoscopic dataset—designed to mitigate domain shifts and serve as a catalyst for advancing endoscopic video processing and its application. Through quantitative analysis, we show that the EndoTracker performs significantly better than the out-of-the-box CoTracker3 foundation model (trained and fine-tuned on computer-vision datasets).

2 Data Curation

Large endoscopic dataset for any point tracking: Our dataset for fine-tuning the foundation model was curated from five endoscopic datasets, carefully chosen to potentially help bridge domain challenges in endoscopic surgery and navigation. These include (i) an in-house sinus endoscopic navigation dataset

generated from 7 cadaveric studies, and four public datasets: (ii) SimCol3D [18] – a synthetic colon dataset, (iii) StereoMIS [12] – dataset for SLAM in endoscopic videos, (iv) Bronchoscopy [6] – benchmarking dataset for bronchoscopy odometry, and (v) PitVis2023 [5] – workflow recognition dataset for pituitary surgery. Given the domain challenges in endoscopic navigation and surgical scenes - such as dynamic scene changes as the endoscope continues to traverse, and partial/full occlusions from instruments/irrigation/blood- that could potentially hinder domain-adapting the point tracking model in a self-supervised setting, the videos from all five datasets were first segmented into short video sequences of 24 frames each. All short sequences were then meticulously filtered by manual review to ensure that at least a partial tissue region remained visible across all frames, ensuring the presence of temporal point correspondences. In total, 13,170 video sequences (comprising > 314k frames) were curated. Fig. 1(a) shows the distribution and samples of the video sequences from the five datasets.

Endoscopic benchmarking datasets: To enable quantitative evaluation of the model’s performance in tracking points in endoscopic scenes, we curate point tracks for three publicly available datasets. (i) **C3VD**: A test set comprising 409 video sequences with ground truth point tracks is generated from a public syn-

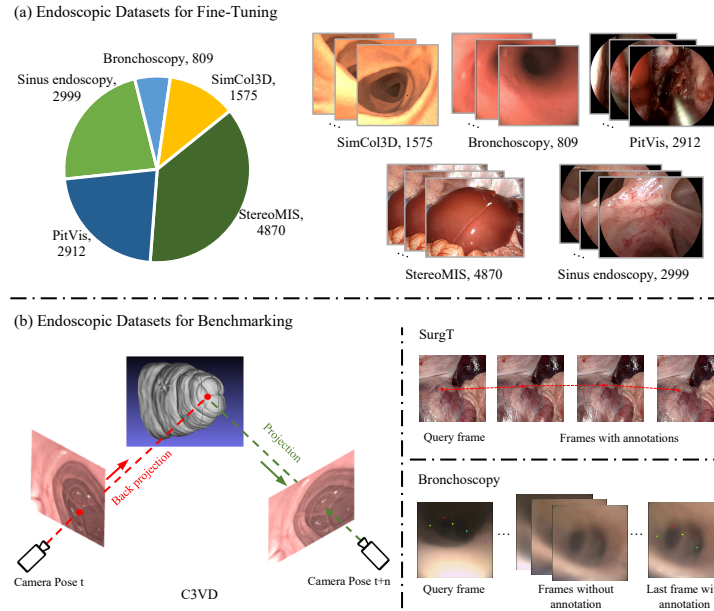


Fig. 1. Dataset overview: (a) Distribution and visualization of our custom-curated endoscopic dataset showing diversity across surgical procedures and tissue types. (b) Construction methodology of our benchmarks (C3VD, SurgT, and Bronchoscopy) utilizing 3D anatomical models and precise camera pose estimation to generate ground truth correspondences for quantitative evaluation.

thetic colonoscopy 3D video dataset [1]. To generate ground truth point tracks across frames, we utilize the 3D model and camera poses from the dataset. We first generate depth maps for camera frames (RGB image) using the 3D model and camera pose. Then, using the depth map and camera pose, a pixel point from one camera frame is first back-projected to a point in the 3D space and then forward-projected to the next camera frame, establishing point correspondences between frames. This process is then iterated for multiple points across 24 frames to generate ground truth point tracks for a video sample. **(ii) SurgT:** A surgical tracking public dataset with bounding box annotation [2]. We use its 125 videos to curate 1768 video sequences as the second held-out test set. To generate ground truth point tracks, we utilize the bounding box annotation, using the bounding box’s center as the ground truth point track across all frames. **(iii) Bronchoscopy:** As bronchoscope data is one of the challenging data to establish point correspondence, we set aside 12 video sequences generated from the Bronchoscopy dataset [6] as the third held-out test set. For this test set, we manually annotated 5 points per sequence, for the first and last frame. Fig. 1(b) illustrates the construction methodology of endoscopic benchmarking datasets.

3 EndoTracker

3.1 Preliminary

CoTracker3 [13] is a transformer-based dense point tracking foundation model. Given query points $P_i^{t_i} = (x_i^{t_i}, y_i^{t_i}) \in \mathbb{R}^2$ at frames t_i for $i \in \{1, 2, \dots, N\}$, the model estimates their positions across all frames. The model is initially trained on synthetic data in a fully supervised manner and then fine-tuned on natural videos in a self-supervised setting employing a teacher-student framework. During inference, the model computes convolutional features per frame and 4D correlations between query point features and all frames. A transformer processes a token grid where for each query point $i = 1, \dots, N$ and time step t , it concatenate bidirectional track embeddings $(\eta_{t-1 \rightarrow t}, \eta_{t \rightarrow t+1})$, confidence C_i^t , visibility V_i^t , and 4D correlations Corr_i^t . Using factorized time attention, group attention across points, and proxy tokens for efficiency, the transformer iteratively refines the tracks, confidence, and visibility through M update steps: $P^{(m+1)} = P^{(m)} + \Delta P^{(m+1)}$, $C^{(m+1)} = C^{(m)} + \Delta C^{(m+1)}$, and $V^{(m+1)} = V^{(m)} + \Delta V^{(m+1)}$. The model offers two modes: an online version for forward-only tracking and an offline version for bidirectional tracking.

3.2 Fine-tuning on Endoscopic Scenes

To domain adapt CoTracker3 to endoscopic scenes, we adopt the original CoTracker3 training regime. In the initial fully-supervised training phase, we leverage the Kubric dataset [10]—a public dataset with comprehensive ground truth point track annotations. Then, extending the model to the endoscopic domain, we fine-tuned the model on our large custom-curated endoscopic dataset in a

self-supervised manner using the teacher-student framework. We use TAPIR [8], CoTracker3-online, CoTracker3-offline trained with window size 12, and CoTracker2 [14] as teacher models. However, unlike the original self-supervised training protocol, where points are first initialized using SIFT descriptors [15], we adopt a uniform grid-based initialization strategy. This choice is motivated by the domain-specific challenges. As the endoscopic scenes often lack distant features on tissue surfaces, the number of points that can be initialized using feature-based methods, such as SIFT descriptors [15], is limited. Therefore, to ensure sufficient points are initialized in the self-supervised training regime, we adopted a uniform grid-based initialization strategy.

4 Experimental Setup

4.1 Datasets

Fully-supervised pre-training dataset: Following the original implementation of CoTracker3, EndoTracker is pre-trained on 6k video samples with ground truth point tracks generated from the Kubric dataset [10]. **Self-supervised fine-tuning dataset:** The pre-trained model is then finetuned on our custom-curated endoscopic datasets (13k video sequences), with pseudo-annotations generated using SOTA point tracking models in a teacher-student framework. These 13k video sequences include all 2999 video sequences generated from the in-house sinus endoscopy dataset, all 1575 video sequences from the SimCol3D, 2912 video sequences from PitVis, 4821 video sequences from StereoMIS, and 795 sequences from the Bronchoscopy dataset. **Test set:** The test set for quantitative analysis includes 409 video sequences generated from C3VD, 1768 sequences from SurgT, and 14 held-out test sequences from the Bronchoscopy dataset. Additionally, 10 video sequences from PitVis and 49 sequences from StereoMIS are used as the held-out test set for qualitative analysis.

4.2 Implementation Details

In this work, we primarily focus on the online mode of the CoTracker3. All models were trained using 8 NVIDIA H100 GPUs on the Johns Hopkins Discovery HPC cluster. The training process consisted of two phases: (1) fully-supervised pre-training on $\sim 6k$ Kubric videos for 200k steps, followed by (2) self-supervised fine-tuning on our $\sim 13k$ custom-curated endoscopic dataset for 60k steps. Across all fully-supervised pre-training regime, we consistently utilized 384 point tracks per video. During fine-tuning, we implemented uniform sampling for query point initialization, maintaining the same density of 384 query points per video sequence. All fine-tuning was conducted with a learning rate of 5×10^{-4} .

5 Results and Discussion

Focusing on the online model, the EndoTracker is benchmarked against out-of-the-box CoTracker2 [14] and CoTracker3 [13] quantitatively on all three of

Table 1. Comparison of EndoTracker against state-of-the-art models in online mode (out of the box), with * denoting statistical significance ($p < 0.05$) of EndoTracker’s performance over CoTracker3. OA: Occlusion Accuracy; C: Confidence.

Model	OA	Jaccard (J)							Points within x pixels (δ_x)							C
		J_1	J_2	J_4	J_8	J_{16}	J_{avg}		δ_1	δ_2	δ_4	δ_8	δ_{16}	δ_{avg}		
C3VD																
CoTracker2	98.41	69.03	88.20	96.36	98.21	98.39	90.04	78.91	93.22	98.61	99.80	99.95	94.10	/		
CoTracker3	99.69	68.41	88.89	97.37	99.38	99.65	90.74	78.41	93.26	98.62	99.77	99.94	94.00	99.93		
EndoTracker	99.53	70.21*	89.64*	97.53*	99.29	99.50	91.23*	79.52*	93.70*	98.79*	99.83*	99.96*	94.36*	99.90*		
SurgT																
CoTracker2	98.31	87.84	97.54	98.10	98.22	98.22	95.99	92.59	99.55	99.94	100.00	100.00	98.41	/		
CoTracker3	98.51	96.93	98.32	98.32	98.32	98.32	98.04	99.04	100.00	100.00	100.00	100.00	99.81	99.57		
EndoTracker	98.38	96.99	98.30	98.30	98.30	98.30	98.04	99.09	100.00	100.00	100.00	100.00	99.82	99.37*		
Bronchoscopy																
CoTracker2	78.59	2.20	2.20	5.48	18.26	35.60	12.75	3.46	3.46	9.62	26.15	45.00	17.54	/		
CoTracker3	69.10	2.06	4.62	8.13	22.64	44.62	16.41	3.46	7.31	12.31	28.08	65.51	23.33	80.39		
EndoTracker	67.56	5.86	8.24	12.46	24.78	45.77	19.42	8.85	14.23	19.62	40.13	63.21	29.21	84.65*		

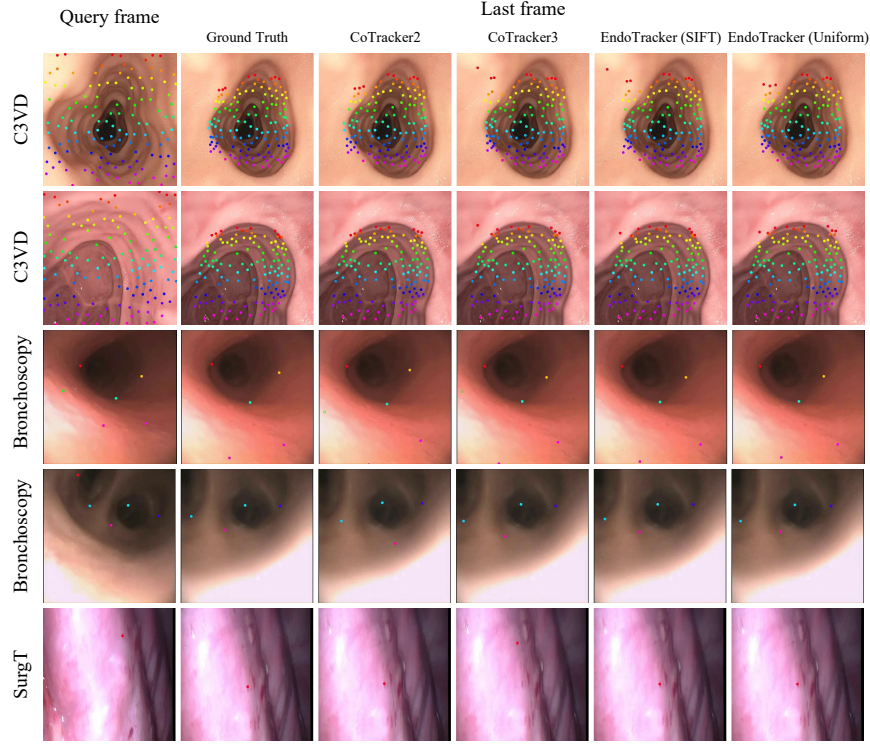


Fig. 2. Qualitative comparison of tracking results on three benchmarks (C3VD, Bronchoscopy, SurgT). The leftmost columns show the initial query points on the reference frame, while subsequent columns show the estimated point positions on the final frame of the sequence.

our test sets (Table 1). While the Jaccard similarity (J) and points within x pixels (δ_x) were calculated across all frames for video sequences curated from C3VD and SurgT, we only calculated their scores based on the last manually annotated frame for the video sequences curated from the Bronchoscopy dataset. Based on the paired t-test, we find that EndoTracker significantly (with statistical significance $p < 0.05$) outperforms out-of-the-box CoTracker3 on the held-out C3VD dataset. Specifically, significant improvement was observed in conservative Jaccard similarity (J_1, J_2 , and J_4) and points within x pixels (δ_1, δ_2 , and δ_4). Achieving better performance in these conservative matrices is crucial for enhancing surgical navigation across challenging anatomical structures. However, no statistically significant improvement in performance is observed in the SurgT or Bronchoscopy dataset. Computationally, the EndoTracker model takes on average ~ 0.08 seconds to inference point-tracks for each window on bronchoscopy dataset, facilitating online inference at ~ 12.5 frames per second.

The lack of significant performance improvement in SurgT dataset could be partly attributed to the presence of clear and distinct visual features present across its scenes, resulting in the out-of-the-box CoTracker3 model achieving very high performance. Additionally, each of its video sequences only has 1 track, reducing the sample size. The general low performance of all models in the bronchoscopy data could be potentially attributed to the extreme scarcity of distinctive visual features on tissue surfaces, which, in our view, necessitates targeted updates to the model architecture in addition to incorporating our domain-bridging datasets.

Fig. 2 shows qualitative comparisons across all evaluated models. On the C3VD dataset, CoTracker2 demonstrates the best overall alignment with ground truth, while EndoTracker shows superior point predictions compared to CoTracker3. In the Bronchoscopy dataset, characterized by fewer and more sparsely distributed tracking points, EndoTracker (uniform sampling) clearly exhibits enhanced position accuracy despite the challenging low-texture, homogeneous appearance of bronchial passages that complicates tracking for all models. For the SurgT dataset, all models except CoTracker3 display comparable performance.

5.1 Ablation Studies

We conducted two ablation studies to evaluate EndoTracker. First, we evaluate query point initialization strategies for tracking performance (Table 2). We observed that EndoTracker trained using a uniform sampling strategy performed significantly better on the C3VD dataset than when trained using a SIFT-based sampling strategy. This could potentially be a result of EndoTracker being fine-tuned on lesser point tracks in the SIFT-based sampling strategy as SIFT may fail to generate a sufficient number of key points in endoscopic scenes due to a lack of distinct features. Uniform sampling forces the EndoTracker to train on a fixed number of samples.

Second, we evaluated the efficacy of domain-specific fine-tuning on our endoscopic dataset (Table 3). EndoTracker consistently outperforms its pre-trained counterparts across all metrics. On the C3VD and Bronchoscopy datasets which

Table 2. Ablation study on query point initialization strategies: Comparison between uniform sampling and SIFT-based keypoint extraction. The symbol * denotes statistical significance ($p < 0.05$) of EndoTracker’s performance when trained using uniform sampling over SIFT-based sampling strategy.

Sampling Strategy	OA	Jaccard (J)						Points within x pixels (δ_x)						C
		J_1	J_2	J_4	J_8	J_{16}	J_{avg}	δ_1	δ_2	δ_4	δ_8	δ_{16}	δ_{avg}	
C3VD														
SIFT	99.34	68.91	88.94	97.16	99.04	99.30	90.67	78.79	93.45	98.70	99.79	99.95	94.14	99.80
Uniform	99.53*	70.21*	89.64*	97.53*	99.29*	99.50*	91.23*	79.52*	93.70*	98.79*	99.83*	99.96*	94.36*	99.90*
SurgT														
SIFT	98.77	97.34	98.51	98.51	98.51	98.51	98.27	99.21	100.00	100.00	100.00	100.00	99.84	99.24
Uniform	98.38	96.99*	98.30	98.30	98.30	98.30	98.04	99.09*	100.00	100.00	100.00	100.00	99.82*	99.37
Bronchoscopy														
SIFT	66.41	2.20	5.43	12.28	27.92	43.24	18.22	3.46	8.46	22.18	43.97	62.44	28.10	81.68
Uniform	67.56	5.86	8.24	12.46	24.78	45.77	19.42	8.85	14.23	19.62	40.13	63.21	29.21	84.65

Table 3. Ablation study on fine-tuning: Performance comparison between our EndoTracker and the original pre-trained CoTracker3. The symbol * denotes statistical significance ($p < 0.05$) of EndoTracker from fine-tuning on endoscopic video sequences.

Sampling Strategy	OA		Jaccard (J)						Points within x pixels (δ_x)							C
	J_1	J_2	J_4	J_8	J_{16}	J_{avg}	δ_1	δ_2	δ_4	δ_8	δ_{16}	δ_{avg}				
C3VD																
CoTracker3 w/o fine-tuning	99.08	57.85	82.5	94.43	98.08	98.87	86.35	70.32	89.15	96.83	99.07	99.7	91.01	99.32		
EndoTracker	99.53*	70.21*	89.64*	97.53*	99.29*	99.50*	91.23*	79.52*	93.70*	98.79*	99.83*	99.96*	94.36*	99.90*		
SurgT																
CoTracker3 w/o fine-tuning	98.83	96.78	97.58	97.63	97.63	98.06	97.54	98.69	99.29	99.34	99.34	99.69	99.27	99.67		
EndoTracker	98.38*	96.99*	98.30*	98.30*	98.30*	98.30*	98.04*	99.09*	100.00*	100.00*	100.00*	100.00*	99.82*	99.37		
Bronchoscopy																
CoTracker3 w/o fine-tuning	66.92	0.85	1.95	1.95	3.88	9.57	3.64	1.54	3.46	3.46	6.92	13.85	5.85	86.26		
EndoTracker	67.56	5.86	8.24*	12.46*	24.78*	45.77*	19.42*	8.85	14.23*	19.62*	40.13*	63.21*	29.21*	84.65		

contain more challenging scenarios, our EndoTracker exhibits notable improvements. This performance gain highlights the quality of our custom-curated fine-tuning dataset and its ability to serve as a bridge between the computer vision and endoscopic domains.

6 Conclusion

Track-Any-Point has the potential to play a catalytic role in transforming endoscopic navigation, surgical automation, and intraoperative decision-making. However, the direct adoption of out-of-the-box computer vision models is potentially hindered by domain-specific challenges in endoscopic scenes. Introducing carefully curated point tracking benchmarking and fine-tuning datasets for endoscopic scenes, this work potentially serves as a bridge in adapting the transformative impact of any point tracking model from computer vision to the endoscopic domain. We also introduce EndoTracker, an adaptation of CoTracker3 for endoscopic scenes. Through extensive qualitative and quantitative analysis, we demonstrate that fine-tuning models on our endoscopic dataset effectively addresses most of the domain shift problem that occurs when models trained

on natural images are applied to endoscopic sequences. Nonetheless, challenges persist in extreme cases—such as bronchoscopy data—where all state-of-the-art models struggle due to the severe lack of distinctive visual features. Our custom-curated endoscopic dataset and EndoTracker serve as a valuable foundation for advancing endoscopic video processing and enabling more robust applications in surgical settings.

7 acknowledgement

This work was funded in part by Johns Hopkins University internal funds and in part by NIH R01EB030511. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

References

1. Bobrow, T.L., Golhar, M., Vijayan, R., Akshintala, V.S., Garcia, J.R., Durr, N.J.: Colonoscopy 3d video dataset with paired depth from 2d-3d registration. *Medical Image Analysis* p. 102956 (2023)
2. Cartucho, J., Weld, A., Tukra, S., Xu, H., Matsuzaki, H., Ishikawa, T., Kwon, M., Jang, Y.E., Kim, K.J., Lee, G., et al.: Surgt challenge: Benchmark of soft-tissue trackers for robotic surgery. *Medical image analysis* **91**, 102985 (2024)
3. Chadebecq, F., Lovat, L.B., Stoyanov, D.: Artificial intelligence and automation in endoscopy and surgery. *Nature Reviews Gastroenterology & Hepatology* **20**(3), 171–182 (2023)
4. Colleoni, E., Edwards, P., Stoyanov, D.: Synthetic and real inputs for tool segmentation in robotic surgery. In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. pp. 700–710. Springer (2020)
5. Das, A., Khan, D.Z., Psychogyios, D., Zhang, Y., Hanrahan, J.G., Vasconcelos, F., Pang, Y., Chen, Z., Wu, J., Zou, X., et al.: Pitvis-2023 challenge: Workflow recognition in videos of endoscopic pituitary surgery. *arXiv preprint arXiv:2409.01184* (2024)
6. Deng, J., Li, P., Dhaliwal, K., Lu, C.X., Khadem, M.: Feature-based visual odometry for bronchoscopy: A dataset and benchmark. In: *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. pp. 6557–6564. IEEE (2023)
7. Doersch, C., Gupta, A., Markeeva, L., Recasens, A., Smaira, L., Aytar, Y., Carreira, J., Zisserman, A., Yang, Y.: TAP-vid: A benchmark for tracking any point in a video. *Advances in Neural Information Processing Systems* **35**, 13610–13626 (2022)
8. Doersch, C., Yang, Y., Vecerik, M., Gokay, D., Gupta, A., Aytar, Y., Carreira, J., Zisserman, A.: TAPIR: Tracking any point with per-frame initialization and temporal refinement. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 10061–10072 (2023)
9. Fujii, R., Hachiuma, R., Kajita, H., Saito, H.: Surgical tool detection in open surgery videos. *Applied Sciences* **12**(20), 10473 (2022)
10. Greff, K., Belletti, F., Beyer, L., Doersch, C., Du, Y., Duckworth, D., Fleet, D.J., Gnanapragasam, D., Golemo, F., Herrmann, C., Kipf, T., Kundu, A., Lagun, D., Laradji, I., Liu, H.T.D., Meyer, H., Miao, Y., Nowrouzezahrai, D., Oztireli, C., Pot,

- E., Radwan, N., Rebain, D., Sabour, S., Sajjadi, M.S.M., Sela, M., Sitzmann, V., Stone, A., Sun, D., Vora, S., Wang, Z., Wu, T., Yi, K.M., Zhong, F., Tagliasacchi, A.: Kubric: a scalable dataset generator (2022)
11. Harley, A.W., Fang, Z., Fragkiadaki, K.: Particle video revisited: Tracking through occlusions using point trajectories. In: ECCV (2022)
12. Hayoz, M., Hahne, C., Gallardo, M., Candinas, D., Kurmann, T., Allan, M., Sznitman, R.: Learning how to robustly estimate camera pose in endoscopic videos. *International journal of computer assisted radiology and surgery* **18**(7), 1185–1192 (2023)
13. Karaev, N., Makarov, I., Wang, J., Neverova, N., Vedaldi, A., Rupprecht, C.: Cotracker3: Simpler and better point tracking by pseudo-labelling real videos (2024), <https://arxiv.org/abs/2410.11831>
14. Karaev, N., Rocco, I., Graham, B., Neverova, N., Vedaldi, A., Rupprecht, C.: Cotracker: It is better to track together (2024), <https://arxiv.org/abs/2307.07635>
15. Lowe, D.G.: Distinctive image features from scale-invariant keypoints. *International journal of computer vision* **60**, 91–110 (2004)
16. Morimoto, T., Kobayashi, T., Hirata, H., Otani, K., Sugimoto, M., Tsukamoto, M., Yoshihara, T., Ueno, M., Mawatari, M.: Xr (extended reality: virtual reality, augmented reality, mixed reality) technology in spine medicine: status quo and quo vadis. *Journal of Clinical Medicine* **11**(2), 470 (2022)
17. Park, J., Park, C.H.: Recognition and prediction of surgical actions based on online robotic tool detection. *IEEE Robotics and Automation Letters* **6**(2), 2365–2372 (2021)
18. Rau, A., Bano, S., Jin, Y., Azagra, P., Morlana, J., Kader, R., Sanderson, E., Matuszewski, B.J., Lee, J.Y., Lee, D.J., et al.: Simcol3d—3d reconstruction during colonoscopy challenge. *Medical Image Analysis* **96**, 103195 (2024)
19. Ravi, N., Gabeur, V., Hu, Y.T., Hu, R., Ryali, C., Ma, T., Khedr, H., Rädle, R., Rolland, C., Gustafson, L., Mintun, E., Pan, J., Alwala, K.V., Carion, N., Wu, C.Y., Girshick, R., Dollár, P., Feichtenhofer, C.: Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714* (2024), <https://arxiv.org/abs/2408.00714>
20. Teufel, T., Shu, H., Soberanis-Mukul, R.D., Mangulabnan, J.E., Sahu, M., Vedula, S.S., Ishii, M., Hager, G., Taylor, R.H., Unberath, M.: Oneslam to map them all: a generalized approach to slam for monocular endoscopic imaging based on tracking any point. *International Journal of Computer Assisted Radiology and Surgery* pp. 1–8 (2024)
21. Tzelnick, S., Rampinelli, V., Sahovaler, A., Franz, L., Chan, H.H., Daly, M.J., Irish, J.C.: Skull-base surgery—a narrative review on current approaches and future developments in surgical navigation. *Journal of Clinical Medicine* **12**(7), 2706 (2023)
22. Zhan, B., Zhao, W., Fang, Y., Du, B., Vasconcelos, F., Stoyanov, D., Elson, D.S., Huang, B.: Tracking everything in robotic-assisted surgery. *arXiv preprint arXiv:2409.19821* (2024)